Quick Note on Lagrangian Mechanics: A Working Treatment for Practical Physicists and Engineers

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1 Variational Calculus

The idea of variational calculus is to find the path that minimizes some sort of functional. But a path is itself a function. So in connection to standard calculus, where we try to find the minimum (or maximum) point of a function, here we try to find the minimizing *function* of a functional.

Consider two fixed points $A(x_1, y_1)$ and $B(x_2, y_2)$, and consider paths $y = f(x)$ connecting them. Depending on what quantity we choose to minimize along the path, we get very different physical problems:

- Minimizing potential energy gives the path of a chain fixed at both ends (the catenary).
- Minimizing distance gives the straight line between A and B.

1.1 Example 1: Minimize Distance

The arc length functional between the two points is

$$I = \int_A^B \mathrm{d}s = \int_{x_1}^{x_2} \sqrt{1 + f'(x)^2}\,\mathrm{d}x.$$

Minimizing I over all functions f recovers the straight line connecting A and B.

1.2 Example 2: Minimize Time — The Brachistochrone Problem

If a particle slides frictionlessly under gravity along a curve $y = f(x)$, the time to traverse it is

$$T = \int_{x_1}^{x_2} \frac{\sqrt{1 + f'(x)^2}}{v}\,\mathrm{d}x = \int_{x_1}^{x_2} \frac{\sqrt{1 + f'(x)^2}}{\sqrt{2g\,(f(x_1) - f(x))}}\,\mathrm{d}x,$$

where v is found from conservation of kinetic and potential energy. Minimizing T over all paths f gives the brachistochrone (a cycloid) — the curve of fastest descent, which is *not* the straight line.

1.3 Example 3: Minimize the Energy — The Catenary Problem

For a hanging rope of linear mass density ρA fixed at both ends, the gravitational potential energy of a segment $\mathrm{d}s$ is $\mathrm{d}(mgh) = \rho A g\, f(x)\,\mathrm{d}s$, so the total potential energy is

$$\Pi = \int_A^B \rho A g\, f(x)\,\mathrm{d}s = \rho A g \int_{x_1}^{x_2} f(x)\sqrt{1 + f'(x)^2}\,\mathrm{d}x,$$

subject to the constraint that the total arc length

$$\ell = \int_{x_1}^{x_2} \sqrt{1 + f'(x)^2}\, \mathrm{d}x$$

is fixed (the rope has a given, unstretchable length). Minimizing Π subject to this constraint gives the shape of a hanging rope or chain: the catenary.

1.4 Example 4: Minimize the Lagrangian

Given $\mathcal{L} = T - V$ (kinetic minus potential energy), a physical system will follow the path that minimizes (more precisely, extremizes) the *action*

$$S = \int_{x_1}^{x_2} \mathcal{L}[x, y, y']\, \mathrm{d}x.$$

Here S is a *functional* — a function of a function. This is the idea we now make precise.

2 Derivation of the Euler–Lagrange Equation

Let $f(x)$ be the optimum path connecting (x_1, y_1) to (x_2, y_2), and let $\eta(x)$ be an arbitrary variation of $f(x)$, subject only to the boundary condition that it vanishes at the endpoints:

$$\eta(x_1) = \eta(x_2) = 0.$$

Consider the family of paths

$$\bar{f}(x) = f(x) + \epsilon\, \eta(x), \qquad \bar{f}'(x) = f'(x) + \epsilon\, \eta'(x),$$

where ϵ is a small parameter.

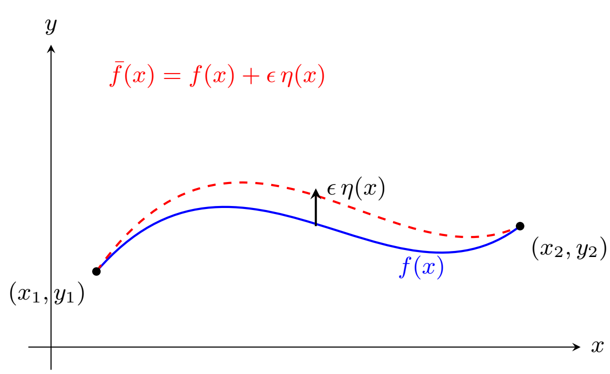

Figure 1: The true extremal path $f(x)$ and a nearby varied path $\bar{f}(x) = f(x) + \epsilon\, \eta(x)$, sharing the same fixed endpoints. The variation $\eta(x)$ is otherwise an arbitrary function, constrained only by $\eta(x_1) = \eta(x_2) = 0$.

Define the functional

$$I = \int_{x_1}^{x_2} F\left[x, \bar{f}(x), \bar{f}'(x)\right] \mathrm{d}x.$$

Once we fix a particular $\eta(x)$, the only thing left that can vary is ϵ — so I is really just an ordinary function of the single number ϵ, exactly like a function you'd differentiate in a calculus class. And since $\epsilon = 0$ gives back the optimal path $f(x)$ itself, $I'(\epsilon)$ must reach its minimum (or maximum) value at $\epsilon = 0$. From ordinary calculus, that means its derivative must vanish there:

$$\left.\frac{\partial I}{\partial \epsilon}\right|_{\epsilon=0} = 0.$$

Carrying out the derivative,

$$\int_{x_1}^{x_2} \left.\frac{\partial}{\partial \epsilon} F\left[x, \bar{f}, \bar{f}'\right]\right|_{\epsilon=0} \mathrm{d}x = 0.$$

By the chain rule (noting $\partial x/\partial \epsilon = 0$),

$$\int_{x_1}^{x_2} \left.\left(\frac{\partial F}{\partial \bar{f}}\frac{\partial \bar{f}}{\partial \epsilon} + \frac{\partial F}{\partial \bar{f}'}\frac{\partial \bar{f}'}{\partial \epsilon}\right)\right|_{\epsilon=0} \mathrm{d}x = 0.$$

Since $\partial \bar{f}/\partial \epsilon = \eta(x)$ and $\partial \bar{f}'/\partial \epsilon = \eta'(x)$, and at $\epsilon = 0$ we have $\bar{f} \to f$, $\bar{f}' \to f'$, this becomes the **first variation** of I:

$$\int_{x_1}^{x_2} \left(\frac{\partial F}{\partial f}\eta + \frac{\partial F}{\partial f'}\eta'\right) \mathrm{d}x = 0.$$

Integrating the second term by parts,

$$\left.\frac{\partial F}{\partial f'}\eta\right|_{x_1}^{x_2} + \int_{x_1}^{x_2} \left(\frac{\partial F}{\partial f} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial F}{\partial f'}\right)\eta \, \mathrm{d}x = 0.$$

The boundary term vanishes because $\eta(x_1) = \eta(x_2) = 0$. Since $\eta(x)$ is otherwise *arbitrary*, the integral can only vanish for all such η if the term in parentheses is identically zero. This gives the **Euler–Lagrange equation**:

$$\boxed{\frac{\partial F}{\partial f} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial F}{\partial f'}\right) = 0.}$$

Example: The Simple Pendulum

Take $F = \mathcal{L} = T - V$ for a pendulum of length ℓ and bob mass m, with θ as the coordinate (so $s = \ell\theta$, $\dot{s} = \ell\dot{\theta}$). Then

$$\mathcal{L} = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell\cos\theta.$$

Applying the Euler–Lagrange equation with θ in place of f and t in place of x,

$$\frac{\partial \mathcal{L}}{\partial \theta} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}}\right) = 0$$

gives

$$-mg\ell\sin\theta - m\ell^2\ddot{\theta} = 0 \quad \Longrightarrow \quad \ddot{\theta} = -\frac{g}{\ell}\sin\theta,$$

the familiar pendulum equation of motion.

Remark 1. *Nothing in the derivation above actually required x to be a spatial coordinate. Had we instead started from a functional $F[t, q(t), \dot{q}(t)]$ — with t as the independent variable and q, $\dot{q} \equiv \mathrm{d}q/\mathrm{d}t$ playing the roles of f and f' — we would have arrived at exactly the same equation,*

$$\frac{\partial F}{\partial q} - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial F}{\partial \dot{q}}\right) = 0,$$

with no change to the derivation itself. Once we identify F with the classical Lagrangian $\mathcal{L} = T - V$, this is precisely the equation of motion of a mechanical system. Section 3 derives, from Newton's second law, why $T - V$ is the right functional to use.

Remark 2. *The derivation also generalizes to functionals depending on higher derivatives,*

$$F = F\left[x, f, f', f'', \ldots, f^{(n)}\right];$$

repeating the same steps (integrating by parts once for every order of derivative present) yields the Euler–Poisson equation

$$\sum_{k=0}^{n} (-1)^k \frac{\mathrm{d}^k}{\mathrm{d}x^k}\left(\frac{\partial F}{\partial f^{(k)}}\right) = 0.$$

For the classical mechanics problems in this note, however, $\mathcal{L}$ depends on q and $\dot{q}$ but never on $\ddot{q}$ or higher, so the first-order Euler–Lagrange equation above is all we will need.

3 d'Alembert's Principle

Starting from Newton's second law for each particle, $\dot{\mathbf{p}}_i = \mathbf{F}_i$, taking the dot product with an arbitrary virtual displacement $\delta\mathbf{r}_i$ and summing over all particles gives the general virtual-work statement

$$\sum_i (\mathbf{F}_i - \dot{\mathbf{p}}_i) \cdot \delta\mathbf{r}_i = 0. \tag{1}$$

This is nothing more than Newton's second law rewritten; no physics has been added yet.

Splitting $\mathbf{F}_i = \mathbf{F}_i^{(a)} + \mathbf{F}_i^{(c)}$ into applied and constraint parts and inserting this split into (1) gives **d'Alembert's principle**:

$$\sum_i \left(\mathbf{F}_i^{(a)} - \dot{\mathbf{p}}_i\right) \cdot \delta\mathbf{r}_i = 0. \tag{2}$$

The constraint forces are gone. They are not neglected — as Examples **??** and **??** show explicitly, they are exactly eliminated, term by term, no matter what their magnitude turns out to be.

4 Lagrange's Derivation from Physical First Principles - Dr. Jorge S. Diaz [3]

Newton's second law as Jakob Hermann stated

$$\sum_k m_k \ddot{\vec{q}}_k = \sum_k \vec{F}_k$$

where q is again our generalized coordinate. To reduce the equation to a scaler we can again use Alembert's principle

$$\sum_k \left(m_k \ddot{\vec{q}}_k - \vec{F}_k\right) \cdot \delta\vec{q}_k = 0$$

Interlude: What does δq actually mean?

It is easy to conflate δq with the ordinary differential dq, or with the time derivative $\dot{q} = dq/dt$ — but all three are different objects, and mixing them up is the single most common source of confusion in this kind of derivation.

Embed the actual path $q(t)$ in a one-parameter family of *nearby, hypothetical* paths

$$q_\epsilon(t) = q(t) + \epsilon\, \eta(t),$$

where $\eta(t)$ is an otherwise arbitrary function (vanishing at the fixed endpoints, $\eta(t_1) = \eta(t_2) = 0$) and ϵ is a parameter that slides between these paths — it is *not* time. The **variation** of q is then defined as

$$\delta q \;\equiv\; \left.\frac{\partial q_\epsilon}{\partial \epsilon}\right|_{\epsilon=0} \;=\; \eta(t).$$

Two consequences follow immediately:

- δq compares different *hypothetical paths* at one fixed instant of time t. It says nothing about how q evolves in t, and is completely unrelated to $\dot{q} = dq/dt$, which instead compares q at two nearby times *along a single fixed path*.
- Because δ (a derivative in ϵ) and d/dt (a derivative in t) differentiate with respect to two independent variables, they commute:

$$\delta\dot{q} \;=\; \frac{\mathrm{d}}{\mathrm{d}t}\,(\delta q)\,, \qquad \text{i.e.} \qquad \left.\frac{\partial}{\partial \epsilon}\left(\frac{\mathrm{d}q_\epsilon}{\mathrm{d}t}\right)\right|_{\epsilon=0} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\left.\frac{\partial q_\epsilon}{\partial \epsilon}\right|_{\epsilon=0}\right).$$

This is exactly the identity being used, silently, every time $\delta\dot{q}$ or $\delta\dot{\vec{q}}$ appears below.

Conservative forces as we know can be taken advantage of using $\vec{F} = -\nabla V$. Using the chain rule

$$\vec{F}\cdot\delta\vec{q} = -\delta V$$

Interlude: Chain Rule

$$\begin{aligned}
\vec{r}_\epsilon &= (x + \epsilon\delta x,\, y + \epsilon\delta y,\, z + \epsilon\delta z),\\
V(\vec{r}_\epsilon) &= V(x + \epsilon\delta x,\, y + \epsilon\delta y,\, z + \epsilon\delta z),\\
\left.\frac{dV}{d\epsilon}\right|_{\epsilon=0} &= \frac{\partial V}{\partial x_\epsilon}\frac{\partial x_\epsilon}{\partial \epsilon} + \frac{\partial V}{\partial y_\epsilon}\frac{\partial y_\epsilon}{\partial \epsilon} + \frac{\partial V}{\partial z_\epsilon}\frac{\partial z_\epsilon}{\partial \epsilon},\\
\delta V &= \frac{\partial V}{\partial x}\,\delta x + \frac{\partial V}{\partial y}\,\delta y + \frac{\partial V}{\partial z}\,\delta z.
\end{aligned}$$

Which can also be obtained directly from

$$\nabla V\cdot\delta\vec{r} = \left(\frac{\partial V}{\partial x}, \frac{\partial V}{\partial y}, \frac{\partial V}{\partial z}\right)\cdot(\delta x, \delta y, \delta z).$$

Thus,

$$\vec{F}\cdot\delta\vec{q} = -\delta V.$$

Therefore, we can instead use the potential in our derivation,

$$\sum_k m_k \ddot{\vec{q}}_k \cdot \delta\vec{q}_k = -\sum_k \delta V$$

which can be further expanded using the identity

$$\ddot{\vec{q}}_k \cdot \delta\vec{q}_k = \frac{d}{dt}(\dot{\vec{q}}_k \cdot \delta\vec{q}_k) - \dot{\vec{q}}_k \cdot \delta\dot{\vec{q}}_k = \frac{d}{dt}(\dot{\vec{q}}_k \cdot \delta\vec{q}_k) - \frac{1}{2}\delta(|\dot{\vec{q}}_k|^2)$$

into the following form that exposes the energy relation,

$$\sum_k m_k \frac{d}{dt}(\dot{\vec{q}}_k \cdot \delta\vec{q}_k) - \underbrace{\sum_k m_k \frac{1}{2}\delta(|\dot{\vec{q}}_k|^2)}_{=\delta T} = -\sum_k \delta V$$

We can move the kinetic term δT to the right and move the derivative out of the summation

$$\frac{d}{dt}\sum_k m_k(\dot{\vec{q}}_k \cdot \delta\vec{q}_k) = \delta(T - V)$$

In order to evaluate the entire continuous path of the function, we must take the integral of both sides, instead of the summation

$$\int_A^B \frac{d}{dt}\sum_k m_k(\dot{\vec{q}}_k \cdot \delta\vec{q}_k)\ dt = \delta\int_A^B (T - V)\ dt$$

the boundary conditions impose the variation to be 0 at the endpoints such that,

$$\vec{r} + \epsilon\delta\vec{r} \to \vec{r}$$

because regardless of the random variation function $\delta\vec{r}$, at the boundaries, the variation must converge to a zero for the path to be defined. Thus using the Fundamental Theorem of Calculus, the left hand side will vanish and reordering gives

$$\delta\int_A^B (T - V)\ dt = 0$$

Thus we know the integrand as the Lagrangian $\mathcal{L} = T - V$. This can be brought to its local form by using first order approximation for the principle of stationary state

$$\delta S = S[q(t) + \delta q] - S[q(t)] = 0$$

Hence

$$S = \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}, t) + \frac{\partial\mathcal{L}}{\partial q}\delta q + \frac{\partial\mathcal{L}}{\partial\dot{q}}\delta\dot{q}\ \ dt - \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}, t)\ dt$$

$$\int_{t_1}^{t_2} \left[\frac{\partial\mathcal{L}}{\partial q} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}}\right)\right]\delta q\ dt + \frac{\partial\mathcal{L}}{\partial\dot{q}}\delta q\Big|_{t_1}^{t_2} = 0$$

Because again at the end points the variation δq is zero, and because δq is arbitrary the integrand must equal to 0. Thus,

$$\frac{\partial\mathcal{L}}{\partial q} - \frac{d}{dt}\left(\frac{\partial\mathcal{L}}{\partial\dot{q}}\right) = 0$$

d'Alembert's principle, in this form, is the standard starting point for essentially every derivation of Lagrangian mechanics found in textbooks: it converts Newton's vector equation into a scalar statement, and it does so *before* we ever have to identify or solve for the constraint forces holding the system together.

5 Generalized Coordinates

Many mechanical systems are subject to constraints: a bead sliding on a wire, a pendulum bob held at fixed radius, the links of a robotic arm connected by pin joints. If a system of N particles is subject to k independent holonomic constraints, only $n = 3N - k$ of the coordinates are actually independent. We collect these into a set of **generalized coordinates** $q_1, \ldots, q_n$ — angles, arc lengths, or any other convenient parameters — such that the position of every particle can be written

$$\mathbf{r}_i = \mathbf{r}_i(q_1, \ldots, q_n, t),$$

with the constraints automatically satisfied for *any* choice of the q_a. Because a virtual displacement is taken at a fixed instant of time ($\delta t = 0$), the chain rule gives

$$\delta \mathbf{r}_i = \sum_{a=1}^{n} \frac{\partial \mathbf{r}_i}{\partial q_a} \, \delta q_a,$$

with no $\partial \mathbf{r}_i / \partial t \, \delta t$ term. The value of generalized coordinates is exactly this: we get to describe the configuration of the system using only as many numbers as it has true degrees of freedom, and the constraint forces never have to be written down at all.

6 Deriving Lagrange's Equations Via Coordinate Transformation and Physics Principles

With d'Alembert's principle (2) established, substitute the generalized-coordinate virtual displacement $\delta \mathbf{r}_i = \sum_a (\partial \mathbf{r}_i / \partial q_a) \, \delta q_a$ and define the **generalized force**

$$Q_a \equiv \sum_i \mathbf{F}_i^{(a)} \cdot \frac{\partial \mathbf{r}_i}{\partial q_a}. \tag{3}$$

Using the identities

$$\frac{\partial \dot{\mathbf{r}}_i}{\partial \dot{q}_a} = \frac{\partial \mathbf{r}_i}{\partial q_a}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathbf{r}_i}{\partial q_a}\right) = \frac{\partial \dot{\mathbf{r}}_i}{\partial q_a}, \tag{4}$$

the inertial term becomes

$$\sum_i \dot{\mathbf{p}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_a} = \sum_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\left(m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_a}\right) - m_i \dot{\mathbf{r}}_i \cdot \frac{\partial \dot{\mathbf{r}}_i}{\partial q_a}\right] = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_a} - \frac{\partial T}{\partial q_a}. \tag{5}$$

Because the q_a are independent, each coefficient of δq_a must vanish separately:

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial T}{\partial \dot{q}_a} - \frac{\partial T}{\partial q_a} = Q_a. \tag{6}$$

This form is worth remembering: it is valid for any applied forces, conservative or not. If the applied forces come from a potential, $Q_a = -\partial V / \partial q_a$, and since V carries no velocity dependence we may fold both terms into

$$\boxed{\mathcal{L} \equiv T - V, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_a}\right) - \frac{\partial \mathcal{L}}{\partial q_a} = 0, \quad a = 1, \ldots, n.} \tag{7}$$

Remark 3. *$\mathcal{L} = T - V$ is a consequence for this class of systems, not a definition, and it fails outside it. The relativistic free particle has $\mathcal{L} = -mc^2\sqrt{1 - v^2/c^2}$, which is not $T - V$ in any useful sense. The defining property of $\mathcal{L}$ is that* (7) *reproduces the correct dynamics.*

7 Worked example: simulating an n-link arm

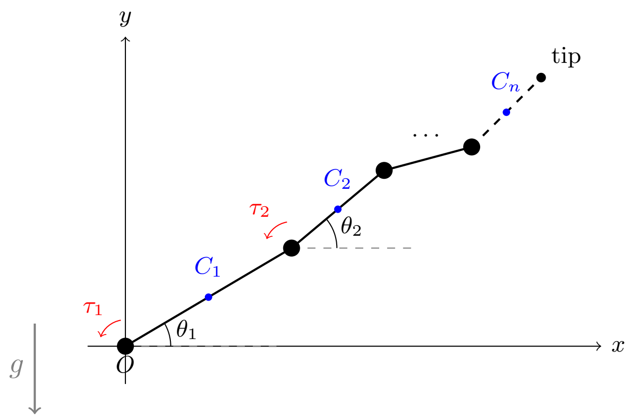

Figure 2: An n-link planar arm hanging in a vertical plane. Link i has mass m_i, length ℓ_i, and moment of inertia I_i about its own center of mass $C_i = (X_i, Y_i)$; θ_i is link i's absolute angle from the x-axis, and a motor at each joint (black) supplies a torque τ_i.

A good stress test for everything above is a small robot arm: n rigid links connected in series by motor-driven pin joints (Figure 2). Unlike the flat-table examples earlier, the arm now hangs in a vertical plane, so gravity does real work on it and the potential energy will not simply vanish.

Take the absolute joint angles $\theta_1, \dots, \theta_n$ (each measured from a fixed horizontal direction) as the generalized coordinates, and let $C_i = (X_i, Y_i)$ be link i's center of mass, taken at its midpoint:

$$X_i = \sum_{j=1}^{i-1} \ell_j \cos\theta_j + \frac{\ell_i}{2}\cos\theta_i, \qquad Y_i = \sum_{j=1}^{i-1} \ell_j \sin\theta_j + \frac{\ell_i}{2}\sin\theta_i. \tag{8}$$

These are the only position expressions needed; everything else is building on top of them.

Step 1: kinetic energy from the velocities. Differentiating gives $\dot{X}_i$ and $\dot{Y}_i$. Once you have those, the kinetic energy is

$$T = \sum_{i=1}^{n} \left[\frac{1}{2} m_i \left(\dot{X}_i^2 + \dot{Y}_i^2 \right) + \frac{1}{2} I_i \dot{\theta}_i^2 \right]. \tag{9}$$

Each link contributes two pieces: its center of mass sliding around (the m_i term) and the link spinning about that center of mass (the I_i term). This step only ever needed velocities — no accelerations appear yet.

Step 2: potential energy from height. Gravity acts in $-y$, and only the height of each center of mass matters:

$$V = \sum_{i=1}^{n} m_i g\, Y_i, \tag{10}$$

reusing the very same Y_i from the position expressions above — no new work required.

Step 3: the Lagrangian and the equations of motion. With T and V in hand,

$$\mathcal{L} = T - V, \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}_i}\right) - \frac{\partial \mathcal{L}}{\partial \theta_i} = \tau_i, \qquad i = 1, \dots, n. \tag{11}$$

This is the point where the accelerations $\ddot{\theta}_i$ actually enter, for the first time in the whole calculation: they fall out of the $\frac{d}{dt}$ in the first term hitting the $\dot{\theta}$-dependence sitting inside T. Nothing before this step needed a single $\ddot{\theta}$.

That is the entire modeling effort, and it goes through with no free-body diagrams and no joint-reaction forces: the pin joints are ideal constraints, and ideal constraints do no virtual work — exactly the property d'Alembert's principle (Section 6) relies on to make the constraint forces disappear — so they are never introduced in the first place.

The result is n coupled, nonlinear, second-order equations in $\theta_1(t), \dots, \theta_n(t)$. Past $n = 2$ these have no clean closed-form solution, and there is no need for one: put in numbers for the m_i, ℓ_i, I_i, and the torque history $\tau_i(t)$, hand the n equations above to Wolfram Alpha (Mathematica), or a Python ODE solver, and let it integrate and plot $\theta_i(t)$ directly.

References

[1] J. R. Taylor, Classical Mechanics, University Science Books (2005). Best first course.

[2] H. Goldstein, C. Poole, J. Safko, Classical Mechanics, 3rd ed., Addison-Wesley (2002). The standard reference.

[3] J. S. Diaz, "Lagrangian Mechanics: When Theoretical Physics Got Real," YouTube, 2025. Available online: https://www.youtube.com/watch?v=QbnkIdw0HJQ.